



Propeller and Propeller-in-Wing Thrust Vectoring

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We investigate the efficiency of a thrust vectoring system utilizing a set of vanes designed to create forward force at a minimum loss in net thrust, placed in both a stand-alone propeller configuration and a propeller-in-wing configuration. Both static and wind-on force-based experiments were conducted at the University of Dayton Low Speed Wind Tunnel (UD-LSWT) with off-the-shelf R/C propellers. A square propeller sheath was incorporated for static testing and a propeller-wing integrated setup was the focus of wind-on experiments. Sensitivity analysis was conducted to determine the effect on thrust vectoring of vane tilt angle and propeller placement with respect to the upper surface of the integrated wing. Static test results indicated notable improvement in vane performance when placing the vane system in a wing as compared to the stand-alone sheathed design. Thrust vectoring was achieved, along with subsequent changes in pitching moment, by increasing vane deflection angle. Wind tunnel test results of the integrated propeller-in-wing system indicated successful thrust vectoring below the advance ratio of 0.3, which is practical for most relevant applications. Sensitivity analysis results revealed that the case of the propeller exposed to the flow freestream outperformed that of the propeller embedded in the test wing in overall efficiency, though the embedded featured a better thrust vectoring capability.

I. Nomenclature

C_D	=	Drag Coefficient	P	=	Power, (W)
C_T	=	Propeller Thrust Coefficient;	Q	=	Torque (Nm)
		$C_T = \frac{F}{\rho n^2 D^4}$	F	=	Thrust (N)
C_Q	=	Propeller Torque Coefficient;	V_∞	=	Freestream Velocity, (m/s)
		$C_Q = \frac{P}{\rho n^3 D^5}$	V_{local}	=	Local Velocity at Propeller, (m/s)
C_P	=	Propeller Power Coefficient;	δ_V	=	Vane Tilt Angle (deg)
		$C_P = 2\pi C_Q$	γ	=	Propeller Pitch, (m)
d	=	Propeller Diameter, (m)	ρ	=	Air Density, (kg/m ³)
D	=	Drag Force (N)			
J	=	Propeller Advance Ratio; $J = \frac{U_\infty}{nD}$			
n	=	Rotational Speed per second			
η	=	Propeller Efficiency/ Efficacy			

Subscripts:

Z	=	Propeller Axial Force
X	=	Propeller Side Force
Y	=	Propeller Normal Force

II. Introduction

Commercial interest in personal air vehicles (PAV) and unmanned air vehicles (UAV) with vertical takeoff and landing capability for air mobility transportation has grown significantly in recent years. Several propulsors have been considered as a primary source of thrust and lift in UAVs including fixed pitch multi-rotors, distributed electric propulsion, and wing-embedded propulsion systems. While the multi-rotor system offers superior hover performance, its efficiency in forward flight is relatively low when compared to fixed-wing aircraft [1-2]. On the other hand, wing-embedded propulsion (WEP) systems or fan-in-wing systems like that of the AgustaWestland Project Zero (Figure 1a [3]) and Ryan XV-5 Vertifan (Figure 1b [4]) have been widely considered as potential alternatives to multi-rotor

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systems which provide higher cruise efficiency. Unlike the multi-rotor system, the WEP system features a greater forward flight efficiency with wings generating much of the lifting force, thus reducing the operational cost thereof [1-2]. Despite the merits of other such competing designs, one major advantage of the multi-rotor system is the added maneuverability it may provide through differential thrust components which are absent in fixed-wing-type UAVs with WEP or distributed electric propulsion (DEP).

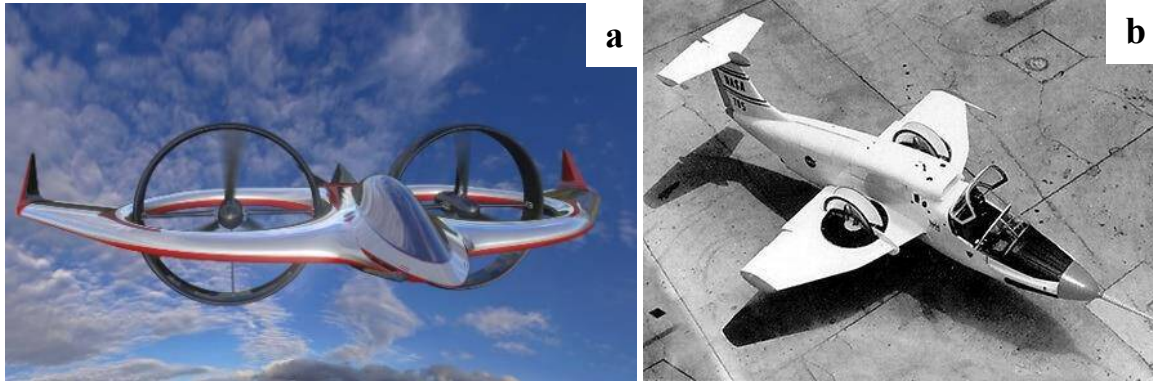


Figure 1. a) AgustaWestland Project Zero [3] with wing embedded propulsion system b) Ryan XV-5 [4] with fan-in-wing propulsion system.

One possible solution to increase the maneuverability of WEP systems for UAVs while maintaining higher cruise efficiency is adding a set of vanes in the wake of the propeller to vector thrust. A set of vanes positioned behind the propeller redirects the high energy wake, thereby achieving thrust vectoring. The vanes may also reduce the swirl component in the wake, increasing thrust efficiency [5-6]. In the mid-1960s, the U.S. Army and General Electric conducted a thorough investigation into state-of-the-art fan-in-wing technology through the Ryan XV-5, a jet-powered single-pilot aircraft. Flight tests examined the experimental design to assess maneuverability, noise footprint, steep terminal approach, and V/STOL performance specifically. Historical reviews by Corgiat et al., Deckert, and Gerdes [7-9] on the XV-5 noted that, though such systems reliably maintain gas-driven lifting, their performance in the transition to forward flight remains non-ideal due to a notably weak low-speed directional control response and complicated power management. This draw-back could be improved upon using more modern flight control systems – which presents the opportunity to revisit and develop a thrust vectoring vane system for a modern PAV and UAV application.

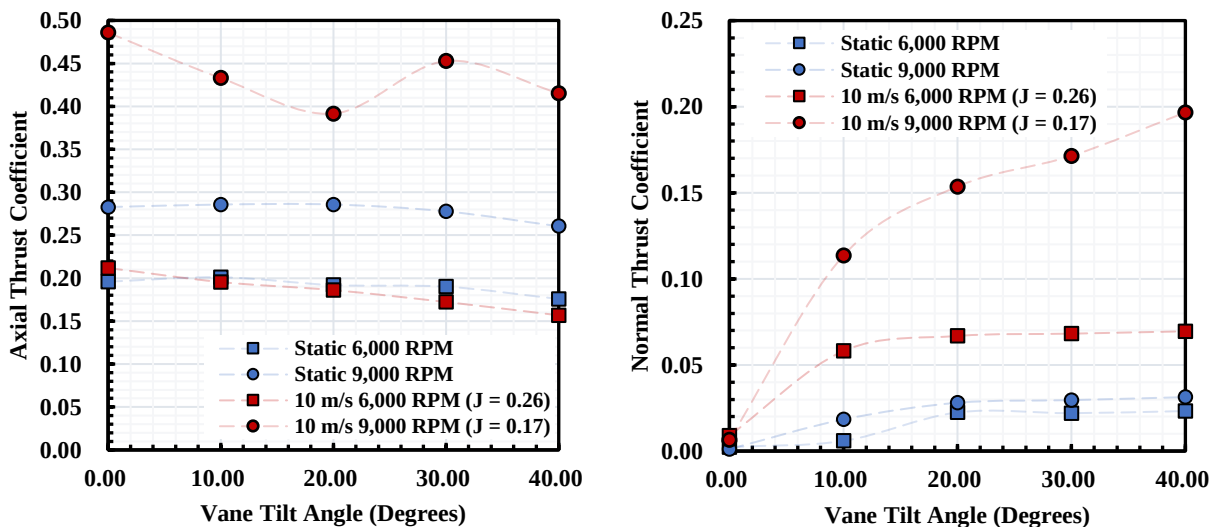


Figure 2. Axial and side thrust coefficient vs. louver tilt angle for the static and wind-on condition. [10]

A study done by Harinarain [10] and Bento [11] investigated two airfoil shaped (NACA 0012) vanes behind the propeller disk, one aligned vertically and the other horizontally, to provide pitch and yaw control. Static experimental results from Harinarain are shown in Figure 2. The propeller disk was placed perpendicular to the freestream. Thus, the axial thrust coefficient C_{Tz} represents the forward thrust performance, and propeller normal thrust coefficient C_{Ty} represents the vectored thrust which provides pitch control. In general, axial thrust C_{Tz} decreases while vectored thrust C_{Ty} increases with the incremental change in vane angle under both static and wind-on conditions. Noticeable increment in the vectored thrust C_{Tx} corresponded to the change of vane tilt angle up to 20° for the static case, and the coefficient remained constant at higher vane angles, likely a byproduct of vane stall. For the wind-on experiment, a higher increment in the vectored thrust coefficient was observed under higher rotations-per-minute (RPM) cases corresponding to a lower advance ratio. These results indicate the potential benefit of using a vane-based thrust vectoring system at small advance ratios, which is the likely operational condition for most UAV and PAV.

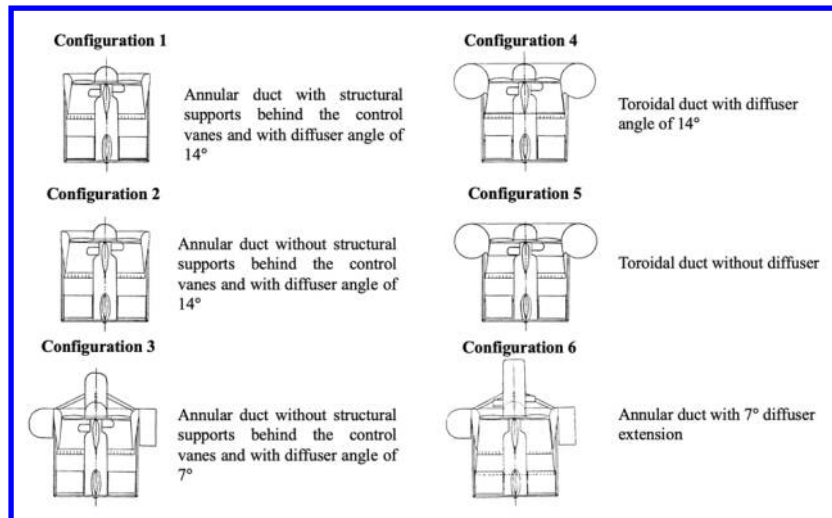
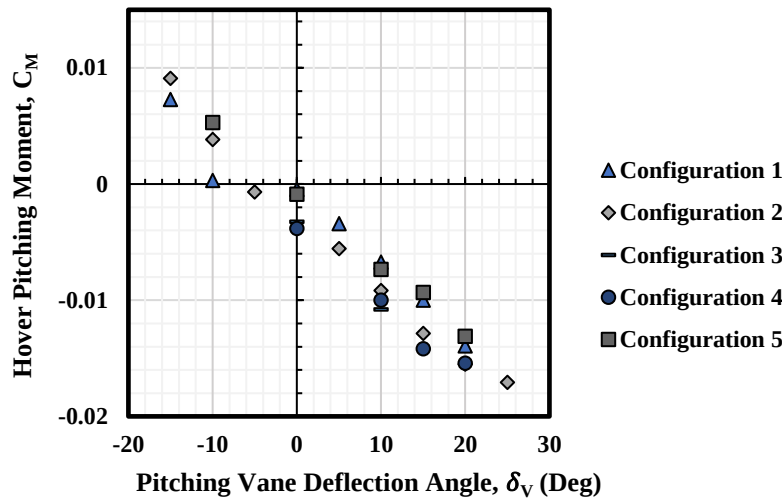


Figure 3. Experimental results showing the connection between pitching vane deflection and hover pitching moment for variable ducted fan configurations studied by Weir. [12]

A study of a free-flying ducted propeller completed by Weir [12] used a vane control system similar to that proposed in the study to provide triaxial stability control. This development of a free-flying ducted propeller system

included wind-tunnel testing of six variable designs under subsonic conditions, measuring in particular the changes in performance due to vane deflection and duct pitch. The designs were placed at variable angles of attack to study the transition performance from hovering to forward flight. Highlighted in Figure 3, six distinct duct configurations were created for the investigation. Configurations 1, 2, and 3 were tied together in basic design, an annular duct with a diffuser; the first had round structural supports downstream of the control vanes, the second did not have these supports so their effect might be quantified, and the third built off of the second with an adjustment in diffuser angle to reduce flow separation. Configurations 4 and 5 included a torus rather than airfoil inlet geometry, the fourth otherwise mimicking configuration 2 to isolate the effects of lip geometry changes and the fifth removing the diffuser feature to assess the importance of the diffuser itself. Configuration 6 was built from the same footprint as configuration 3 with an extension to the diffuser to shield the vanes in the duct from effects of jet bending and stall. Each of the six configurations, with different inlet duct geometries, fan support structures, and exit diffuser designs incorporated a set of NACA 0018 pitching vanes placed in the propeller wake. Figure 3 above shows the experimental results of vane deflection on the hover pitching moment of the system. The results indicate a high effectiveness of the control vanes in influencing the system's pitching moment. Moreover, the linear trend of changes in pitching moment indicates a higher vane airfoil stall angle when compared to the usual airfoil stall angle for fixed wing study.

Deckert [8] examined different WEP design configurations under static and dynamic tests. Model 2 (annotated in Figure 4) includes a larger fan integrated in the wing for lift production while Models 4 and 6 have smaller fans integrated which provide roll and pitch moment, respectively. In Figure 4, the fan-in-wing induced lift, L_i/T_s , which is the ratio of the lift induced by the integrated fan to the total thrust generated by the fan, is shown for a range of advance ratios. Advance ratio was changed in this case by adjusting the airspeed of the tested system. In most of the cases, placing a rotor in the wing resulted in a positive effect of the lift generation, except for the Model 6c, which placed the fan closer to the leading edge of the wing causing a reduction of the flow speed over the upper wing surface.

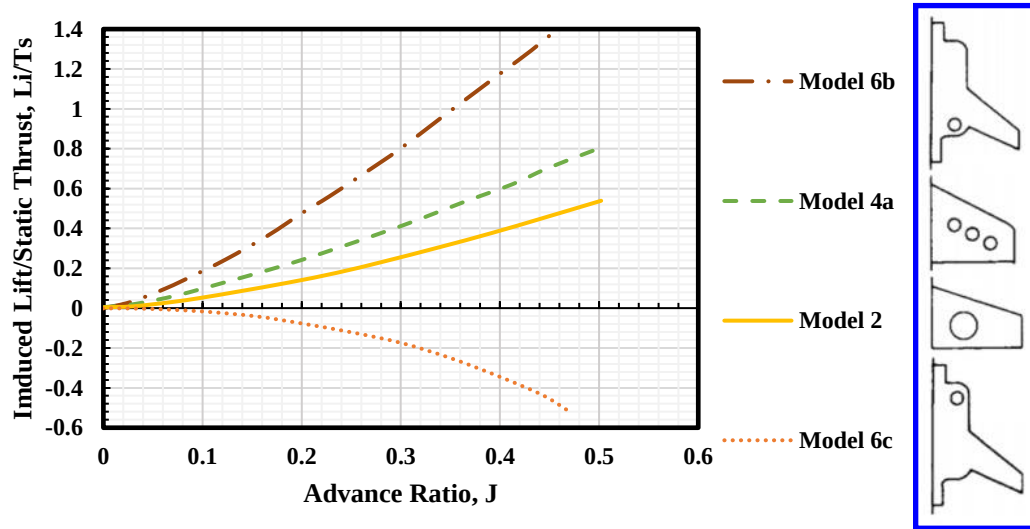


Figure 4. Induced lift variation with advance ratio for a set of fan-in-wing designs as published by Deckert. [8]

Hickey and Ellis [13] investigated the performance of a semi-span wing featuring an embedded fan with both inlet and exit vanes acting independently as shown in Figure 5. A NACA 16-015 wing profile was used for the wing of 12.5 square foot area, and a duct of 1.63 foot-diameter was placed inside the wing's 2-foot chord. All 15 exit vanes of the fan duct featured a NACA 65-010 airfoil profile. Wind tunnel tests maintained a constant wind tunnel pressure, q_∞ , and propeller speeds were varied between 0-8000 RPM. In this study, it was concluded that exit vanes could be used to generate a net propulsive force for the attached system in a range of freestream velocities with only a small reduction in lift. Inlet vanes, by contrast, caused no measurable change in static lift when individually manipulated to maintain flow attachment; when all inlet vanes were axially aligned, static lift was increased – along with power requirements. Thus, though inlet vanes did show potential in reducing nonuniform propeller inflow, each vane would

need to be independently adjusted with forward speed to provide a performance gain rather than a loss. Exit vanes were then considered to be a viable option for thrust generation in forward flight as they reduce static lift and increase power slightly under such conditions. Figure 6 below highlights published results of the study where the following equations define each coefficient:

$$C_F = \frac{L_{propeller}}{q_{\infty} S} \quad (1)$$

$$C_M = \frac{M}{q_{\infty} S} \quad (2)$$

$$C_L = \frac{L_{total\ wing}}{q_{\infty} S} \quad (3)$$

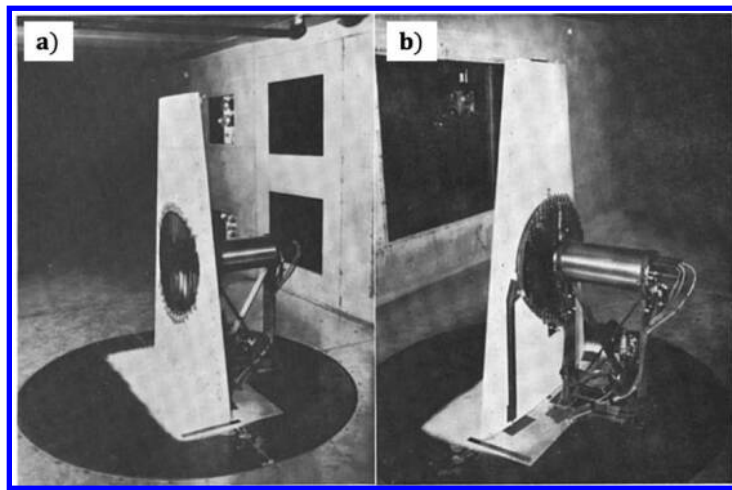


Figure 5. Fan-in-wing model studied by Hickey and Ellis with a) inlet vanes and b) exit vanes. [13]

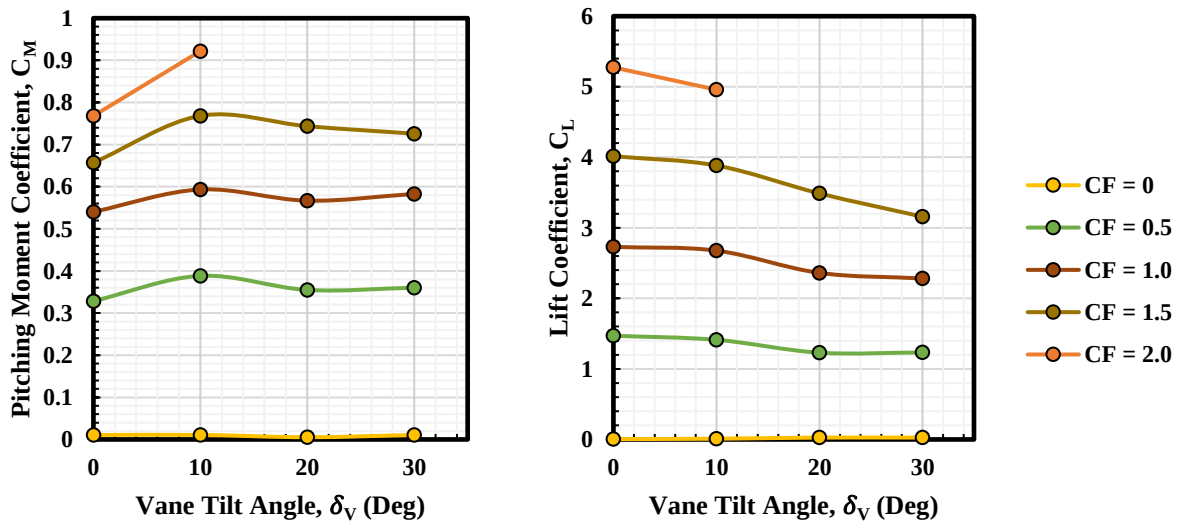


Figure 6. Relationship between exit vane deflection angle and a) pitching moment coefficient and b) total lift coefficient for integrated wing design with respect to constant propeller force as studied by Hickey and Ellis. [13]

This review of literature provides reasonable grounds for confidence in effective thrust vectoring from exit vanes as well as their ability to influence pitching moments at relatively low expense to the lift or static thrust for the system in which they are set. Historically, most of the literature mentioned utilized a lifting fan over a propeller. Due to increased use of R/C and sport propellers in UAVs and air taxis, we perform investigations to study the effect of thrust vectoring vanes on a stand-alone propeller system as well as the propeller-in-wing system.

III. Experimental Setup

A. General Experimental Setup

Experiments were conducted on a thrust-stand at the University of Dayton Low Speed Wind Tunnel Lab (UD-LSWT). The APC 11-inch by 7-inch thin electric propeller series was selected for testing because of the abundance of available results published with the same design selection from comparable literature data references. The propeller was driven by an OMA-3810-1050 brushless out-runner electric motor, itself powered by a PSW 30-108 constant-voltage supply. The peak Reynolds number of the propeller blades is 78,000, which was calculated based on the blade chord at 70% blade-semi-span and the peak propeller rotational speed achieved. The propeller rotational speed range, with its lower bound derived from signal-to-noise considerations and upper bound limited by the speed control capacity, was 60-120 RPS.

The propeller and motor were attached to an ATI Industrial Automation Mini-40 six-component force balance (www.ati-ia.com) [14]. Axial thrust is the force along the Z-axis of the sensor, while the vectored thrust is the force along the Y-axis of the sensor; the propeller torque is then the rotational moment about the Z-axis. The balance bolts to an aluminum-extrusion frame secured to the lab floor and wind tunnel contraction. The balance has a maximum of 120-Newton (N) force on its Z-axis and maximum torque of 2 Nm about its Z-axis. The encountered peak thrust and torque were 12 N and 0.25 Nm respectively. The data sampling rate was maintained at 1,000 Hz for a 15-second duration for each testing condition with two runs per each to improve data repeatability. In an effort to minimize sensor drift, tare values were taken before and after each test. The average of the two tares was then subtracted from the respective thrust and torque readings to correct for drift. A band-pass filter was used for the data post-processing, filtering out the noisy frequencies related to propeller rotational speed and thrust stand natural frequency.

a.) Standalone Propeller-Vane Thrust Vectoring Test Setup

For static testing of the standalone propeller-vane setup, four 7.62 centimeter (cm)-chord vanes made with NACA 0012 airfoil sections were utilized. The fundamental relationship between the vane deflection and vectored thrust is shown in Figure 7a. The vane tilt angle δ_v was locked in place by a rigid fixture, and the fixture was physically altered each time a different vane tilt angle is tested. The tip of each vane connected to a square duct sheath (shown in Figure 7b) to eliminate vane tip vortices. The square duct-vane setup was then bolted on 80-20 aluminum extrusion frame, itself secured to the lab floor and test section inlet. The rotational speed of the propeller was measured by a laser mounted at the ceiling of the lab communicating with a photodiode mounted beneath the propeller-vane setup. The propeller-vane setup was placed at least one propeller diameter away from any type of blockage to minimize the error related to experimental setup [15].

Force-based testing was conducted using two different standalone propeller-vane-sensor interface configurations. The first configuration was designed to measure the overall force experienced by the propeller-vane system. The second configuration was designed to only measure the forces experienced by the propeller in order to quantify the effect of the vanes on propeller performance, especially the power/torque coefficient. The second configuration was justified by the diminishing of the overall torque of the propeller in the presence of the vectoring vanes, resulting in a power coefficient order of magnitude lower for the propeller-vane system when compared to the bare propeller. The schematics of the two different configurations are shown in Figure 8. One of the main differences between the two configurations was the strategic placement of the force sensor to measure the forces in overall propeller-vane system in the first configuration and the propeller alone in the second.

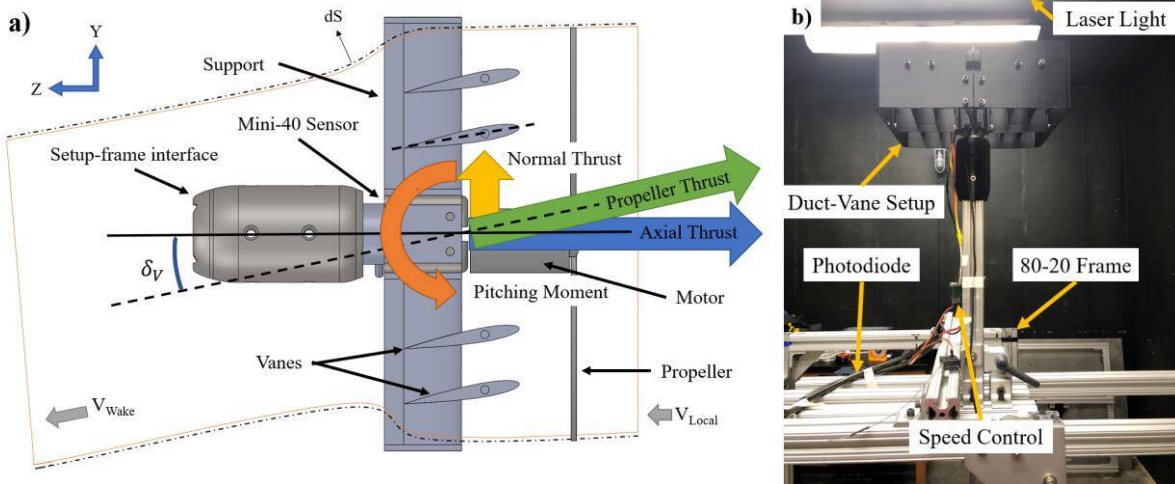


Figure 7. a) Sketch of propeller duct-vane test setup indicating the relationship between force components and propeller wake. b) Experimental setup of the square duct-vane propulsion system for static testing.

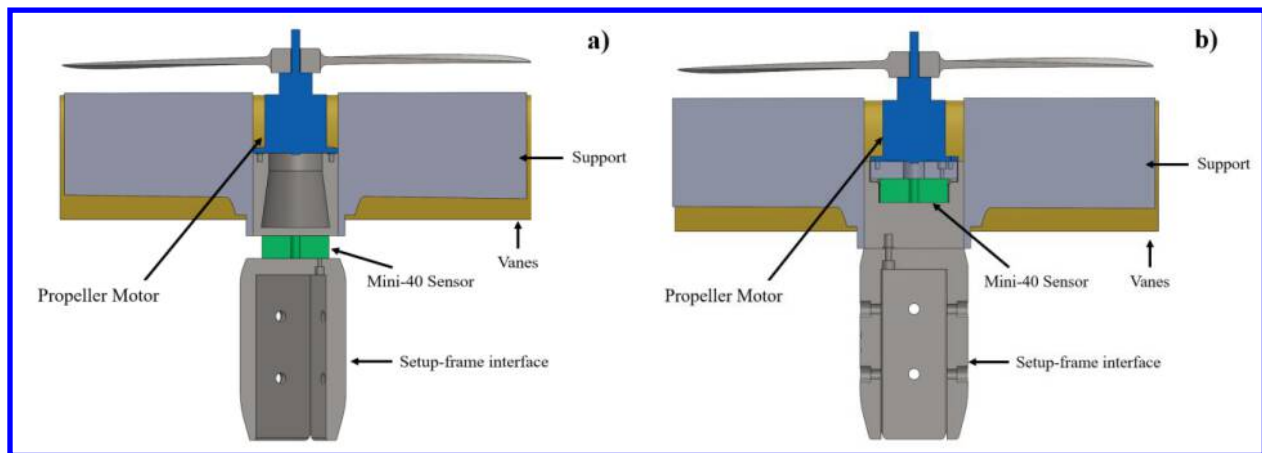


Figure 8. Section view of the force-based testing configurations to measure forces on a) propeller-vane assembly b) propeller only in the propeller-vane assembly.

b.) Propeller-In-Wing Thrust Vectoring Test Setup

The propeller-in-wing experiments were conducted at the UD-LSWT with the tunnel in the open jet configuration. The UD-LSWT has a 16:1 contraction ratio, 6 anti-turbulence screens, and 76.2 cm by 76.2 cm open jet test sections. The effective length of the test section is 182.9 cm. A 111.8 cm by 111.8 cm collector accepts the expanded air on its return to the diffuser. The freestream turbulence intensity is 0.1% at 15 meters per second (as measured by hot-wire anemometry). A pitot tube connected to a TSI T600 Micromanometer was used to measure the freestream velocity during the experiment.

The standalone propeller-vane thrust vectoring test setup shown in the previous section was housed inside a wing with a span of 83.32 cm and a chord length of 60.96 cm. This setup will be mentioned as propeller-in-wing system for the remainder of this paper. The wing section was constructed using an Eppler 479 airfoil, sufficiently thick to properly house the propeller-vane system. The wing was constructed from a series of wooden ribs with an Eppler 479 profile cut with laser cutter. These ribs were connected by a leading edge and trailing edge spar made of 80-20 aluminum rail. The rib and spar structure of the wing was later covered with a MonoKote [16] which formed the skin of the wing. The internal assembly of the propeller-in-wing test setup along with the fully integrated test setup is shown in Figure 9. The wing was then secured to the test section inlet (as shown in Figure 9c). A square opening with a length of 29.21 cm was designed to house the propeller-vane system. The center of the propeller and the balance

sensors were placed at maximum thickness/chord location (35% chord), which has been shown to provide a better performance in the wing lift generation. [17]. The mock wing mounted to the aluminum extrusion frame which was then secured on the tunnel contraction to be isolated from the propeller-vane propulsion system. The wing was designed to be completely isolated from the propeller-vane system so that force-based testing only measures the forces on the propeller-vane system and not the propeller-in-wing system as a whole.

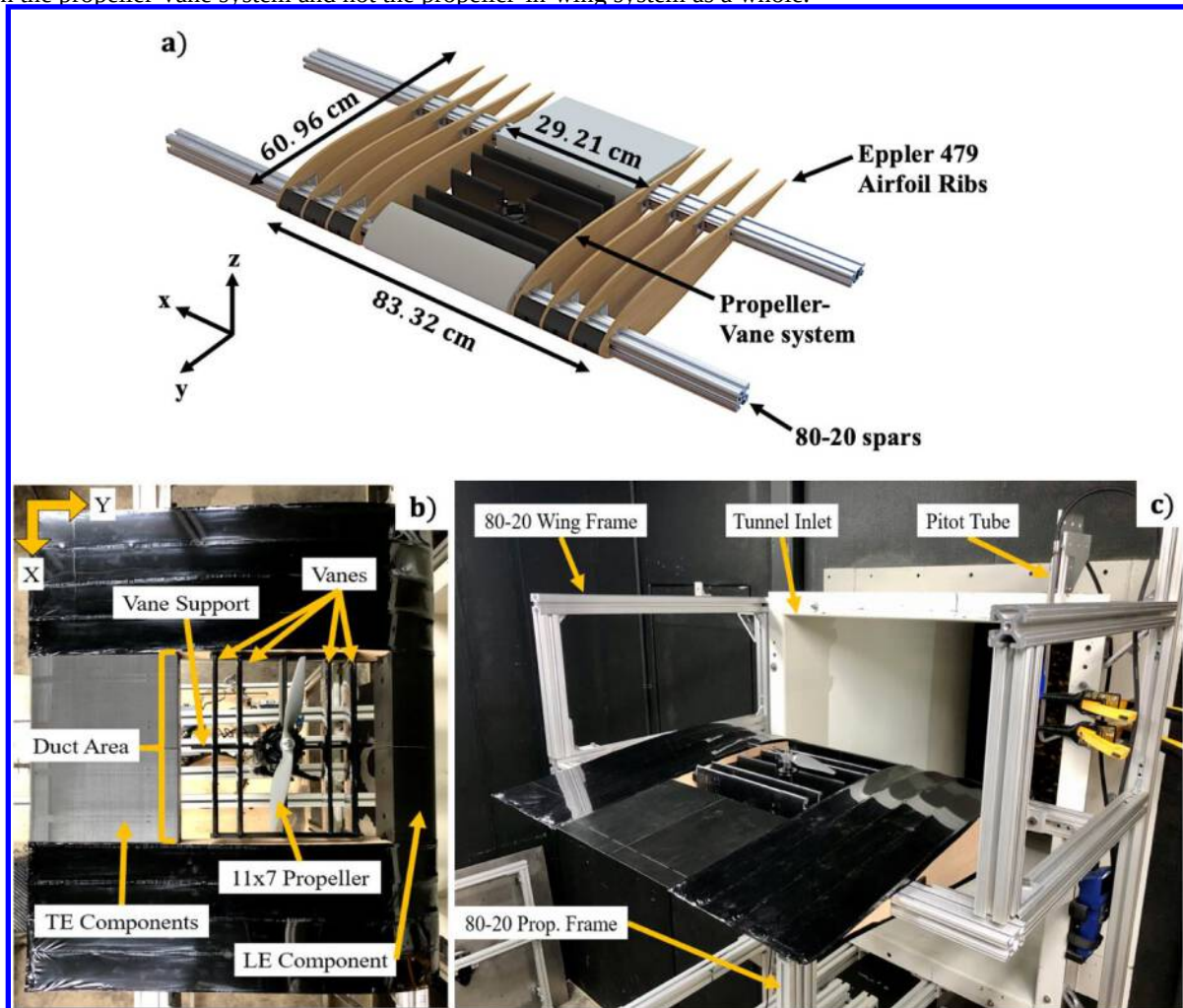


Figure 9. a) Internal assembly of propeller-in-wing test setup. b) Aerial view of mounted propeller-in-wing setup and c) isometric view of propeller-in-wing test setup.

When compared to the vane chord length used in thrust vectoring studies mentioned in the introduction section (an average of 9% of the propeller diameter), the current study utilizes a longer vane chord length (28% of the propeller diameter) to increase the vane thrust vectoring performance. As such, judicious decisions were considered to expose either the vanes or the propeller to the freestream from within the wing opening, a practical conundrum that a designer would face given constraints on the wing thickness. To assess the differences in system performance between propeller-vane placement cases, we investigate the propeller-vane performance in the propeller-in-wing system when the propeller is embedded in the wing (vanes exposed to the freestream) as well as when the propeller is exposed to the freestream (vanes embedded in the wing). These two configurations are shown in Figure 10.

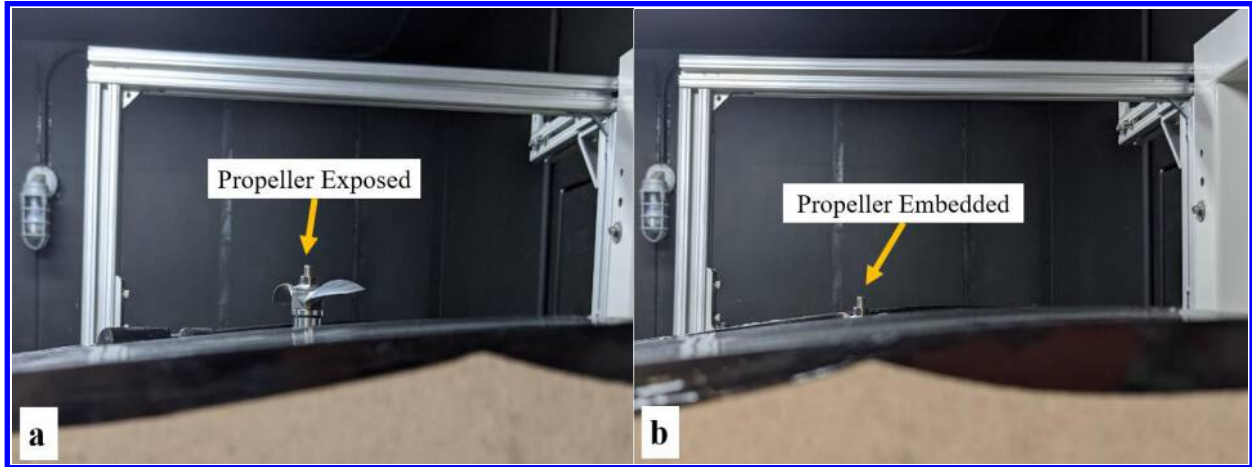


Figure 10. a) Side view of exposed propeller and b) side view of embedded propeller.

B. Test Matrix

a) Static testing

Static tests were conducted to examine the performance of the standalone propeller-vane system as well as the propeller-vane system embedded in the wing (propeller-in-wing system). A bare propeller was also tested to serve as a baseline for the static experiments. An APC 11-inch by 7-inch propeller was used for all the experiments at a rotational speed range from 60-120 RPS and a peak blade Reynolds number of 78,000. Six different vane tilt angles were tested for all configurations, and 2 trials of each experimental condition were completed for a repeatability check. Table 1 below summarizes the static tests conducted for each of the three test iterations.

Table 1. Static Test Matrix Including All Test Iterations

Test Cases	Propeller	Propeller Rotational Speed (RPS)	Propeller Reynolds Number	Vane Tilt Angle δ_v (degrees)	Trials
Bare Propeller	APC 11x7	60-120	35,000-78,000	N/A	2
Standalone propeller-vane system				0, 5, 10, 15, 20, 30	
Propeller-in-wing system (Embedded and Exposed)				0, 5, 10, 15, 20, 30	

To verify the reliability of the propeller test stand, bare propeller static testing results obtained from the setup mentioned in this section were compared to the UIUC database [18] as well as two previous University of Dayton Low Speed Wind Tunnel (UD-LSWT) studies conducted by Cai et al. [19] & Gunasekaran et al. [20]. The comparison of the thrust and power coefficient is shown in Figure 11. The error bars represent a 95% confidence interval for the propeller results after noise filtration. For both thrust coefficient and power coefficient, current experimental data is comparable experimental result of both UD-LSWT studies and UIUC database, which provides confidence in the

current experimental setup. The results indicate a slight increment of C_T with propeller rotational speed due to the Reynolds number effect while the C_p remains relatively constant across the RPM range tested.

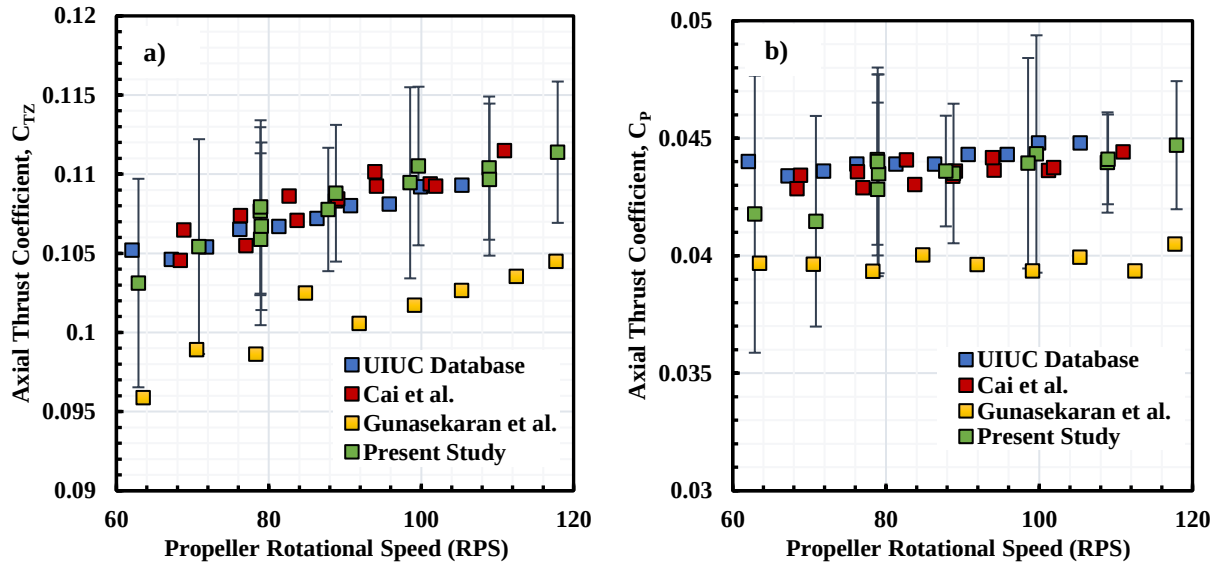


Figure 11. a) Thrust coefficient and b) power coefficient comparison for APC 11x7 propeller.

b) Wind-on Propeller-in-Wing Testing

The test matrix for the wind-on propeller-in-wing tests is shown in Table 2. Two propeller elevations were tested with fixed propeller rotational speed. Different advance ratios were obtained by changing the freestream velocity in the wind tunnel from 0 – 19 m/s. The motor power output was adjusted as necessary to maintain a constant propeller rotational speed across tunnel speed conditions. Two trials were completed to check for data repeatability.

Table 2. Dynamic Test Matrix Including All Test Iterations

Test Case	Propeller Location	Propeller Rotational Speed (RPS)	Advance Ratio, J	Wind Tunnel Speed (m/s)	Vane Tilt Angle δ_v (degrees)	Sensor Interface
Bare Propeller	N/A	100	0 - 0.7	0 - 19	N/A	Propeller only (Figure 8a)
Propeller-In-Wing system	Embedded				0, 10, 20, 30	Propeller only (Figure 8a);
	Exposed					Propeller-vane system (Figure 8b)

IV. Results

The experimental results for thrust vectoring are split into two sections: The first section consists of the results from the static testing of the standalone propeller-vane system and the propeller-in-wing system. The results from this section show the efficacy of the thrust vectoring during takeoff and transition to forward flight. The second section consists of the results from the wind-on testing of propeller-in-wing system. The results from this section show the efficiency of the thrust vectoring system during forward flight at different advance ratios.

A. Static Testing Results

Experimental results for axial thrust coefficient and power coefficient are shown below in Figure 12 for the bare propeller, standalone propeller-vane system (SA-PV), and the propeller-in-wing system (PVW) with propeller

embedded and exposed. The vanes were set to a zero-degree deflection ($\delta_v = 0^\circ$) for the results shown in Figure 12. The differences between the bare propeller and the standalone propeller-vane system describe the effect of vanes on the propeller performance; the differences between the standalone propeller-vane system and the propeller-in-wing system indicate the effect of the wing enclosure on the propeller performance. It should be noted that the results shown in Figure 12 were obtained using the second configuration of the propeller-vane-sensor interface shown in Figure 8 where only the forces experienced by the propeller are measured. The results indicate a reduction of total axial thrust for the propeller-vane system of 4% due to the drag force on the vane and an additional 6% of loss in axial thrust due to the wing enclosure effects. In terms of power coefficient, the change in power consumption became negligible when placing the vane system in the propeller wake. An average incremental difference of 3% was observed between the stand-alone and integrated propeller-vane systems due to the wing interference when placing the system therein.

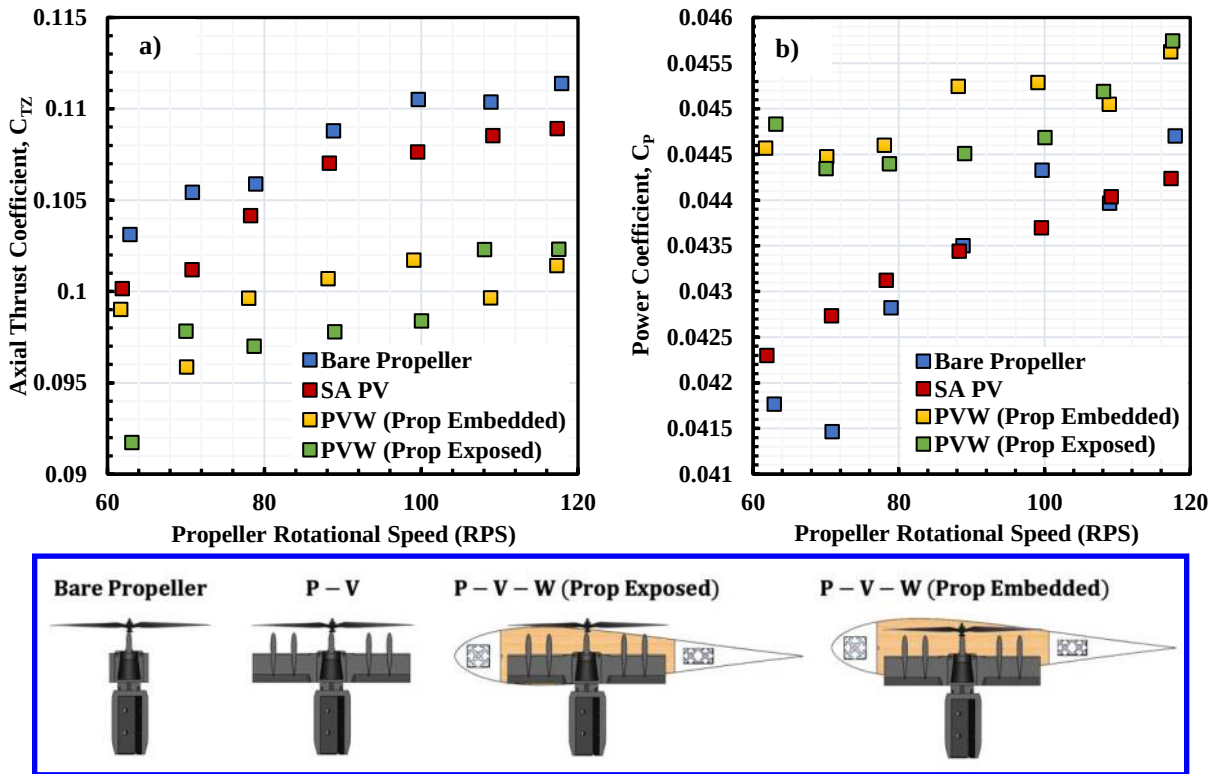


Figure 12. Comparing propeller rotational speed and a) axial thrust coefficient and b) power coefficient across iterations when $\delta_v = 0^\circ$.

The normal force and pitching moment generated by the propeller-vane system at different vane deflection angles are shown in Figure 13. An increment in normal force coefficient C_{Ty} with higher vane tilt angle is observed for both standalone propeller-vane system and the propeller-in-wing system. The increment in C_{Ty} for both PVW systems with respect to the vane tilt angle δ_v is approximately linear with a significantly improved performance as compared with the SA-PV system. Moreover, the curve slope of C_{Ty} reduces between $\delta_v = 20^\circ$ and $\delta_v = 30^\circ$ cases for the SAPV system, likely a result of vane stall. Both phenomena indicate a positive effect from wing interference for vectored thrust. Additionally, the propeller placement in the integrated wing seems to have a relatively minor effect on the normal force coefficient.

Figure 13b shows the resultant pitching moment coefficient C_{Qx} across variable vane deflection angles for each static test case. A negative pitching moment coefficient indicates a stabilizing, pitch-down moment on the wing. An increment in the magnitude of C_{Qx} occurs with incremental change of the vane tilt angle δ_v . Similar to the vectored

thrust results, the integrated wing has a positive effect on the generation of C_{Qx} by the vane system. Again, the propeller location seems have minor effect on C_{Qx} .

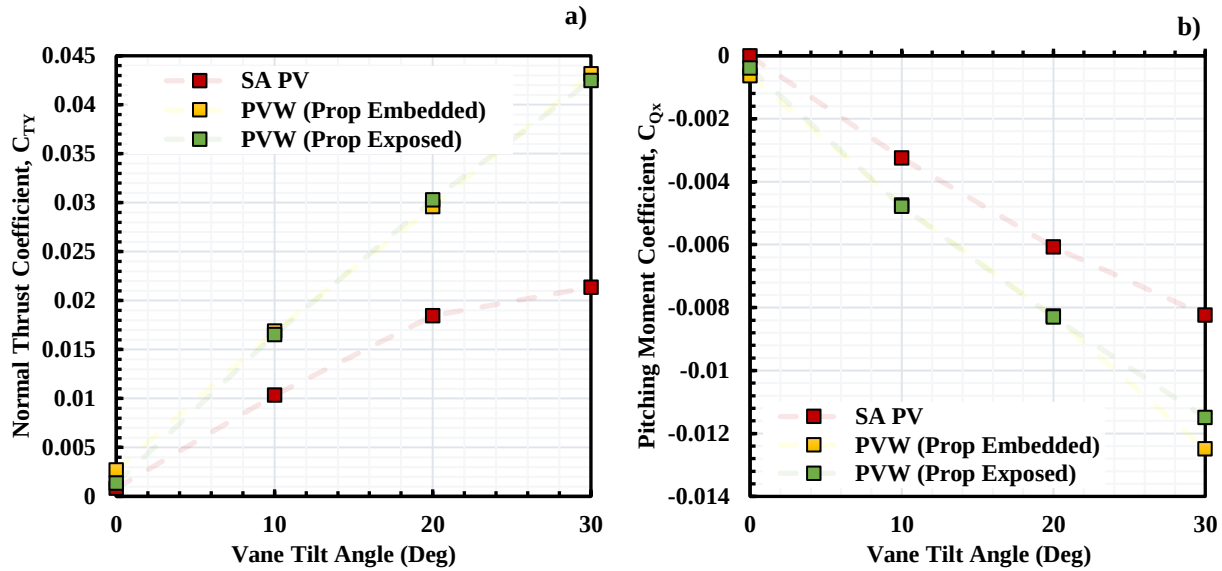


Figure 13. a) Normal force coefficient and b) pitching moment coefficient with respect to changing vane tilt angles.

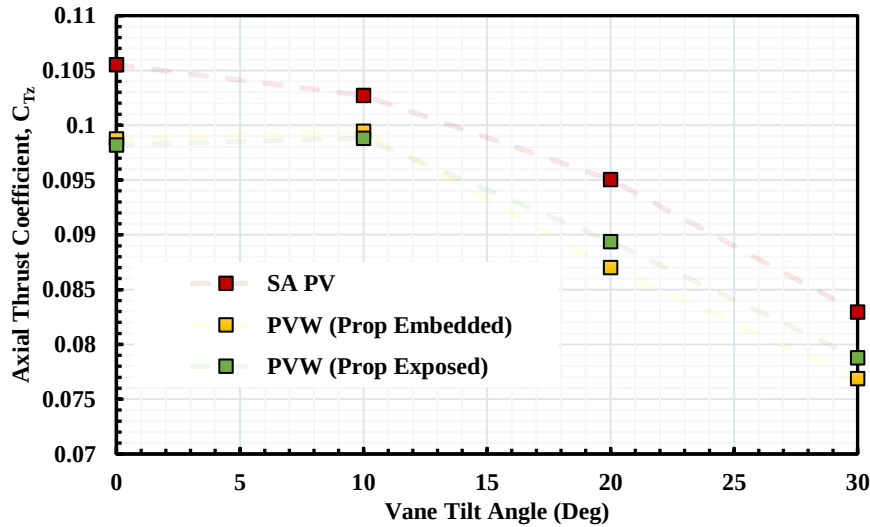


Figure 14. Propeller axial thrust coefficient vs. RPS across different vane tilt angles.

Axial thrust coefficient results, which speak to the system's vertical takeoff capability, across different vane tilt angles are shown in Figure 14. Generally, the axial thrust force decreases with the increment of vane deflection angle δ_v . For the PVW systems, there is little difference between the $\delta_v = 0^\circ$ and $\delta_v = 10^\circ$ cases, while an average difference of 3.5% exists between $\delta_v = 20^\circ$ and $\delta_v = 30^\circ$ for the exposed propeller case which experiences greater axial thrust losses than the embedded propeller case. Compared to the SA-PV system results, the wing interference seems to have a negative effect on the axial thrust generation with a 6% average loss of axial thrust observed. However, it is worth noting that the decrease in axial thrust coefficient for the integrated wing correlates to a stronger thrust vectoring performance. These results indicate a minor effect from the wing interference to the overall efficiency of the propeller-vane system. The efficacy of the propeller-vane system is determined by:

$$\eta_\delta = \eta_\theta * \eta_T \quad (4)$$

where:

$$\eta_\theta = \frac{\tan^{-1}\left(\frac{T_Y}{T_Z}\right)}{\delta_v} \quad \text{and} \quad \eta_T = \frac{\sqrt{C_{Ty}^2 + C_{Tz}^2}}{C_{Tz \text{ Bare Prop}}} \quad (5)$$

Equation 4 describes the ability of the vanes to deflect the propeller wake without losses in net thrust. Under ideal conditions, if the wake deflection is equal to the vane angle, η_θ will be equal to one. If there no losses in the net thrust, then η_T will be equal to 1. All efficacies are plotted together in Figure 15 below.

The overall wake momentum vectoring efficiency η_δ decreases with the incremental change in vane tilt angle δ_v . For both PVW systems, the value of η_θ remains constant with the increment of vane deflection angle δ_v , around 0.95, while the value of η_θ for the square sheath case reduces with the increment of δ_v . On the other hand, the value of η_T for all three systems is similar with a 10% decrement observed from $\delta_v = 10^\circ$ to $\delta_v = 30^\circ$. The resultant η_δ for the standalone propeller-vane system is significantly lower than that of both PVW systems as a result of very low η_θ conditions, also observed in the normal force coefficient results plotted in Figure 13. At the highest vane tilt angle, both PVW systems maintain about 80% efficacy with respect to thrust vectoring. This provides confidence for propeller pitching performance with respect to the wing system, improving forward flight performance at cruise while still generating enough vertical thrust for vertical takeoff and landing operations in hover.

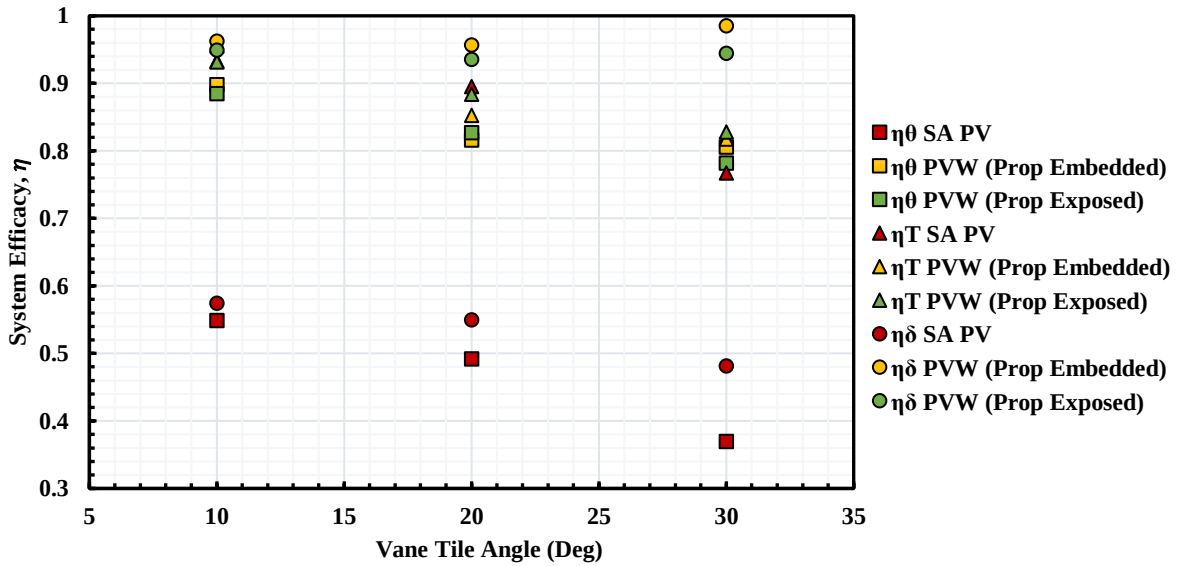


Figure 15. η_θ , η_T and η_δ with respect to vane tilt angle for all configurations under hover condition.

B. Wind-on Testing Results

The propeller rotational speed was fixed during the wind-on experiment while the incoming freestream velocity was changed to obtain different advance ratio conditions. It is worth noting that the propeller disk was placed parallel to the freestream, resembling edgewise flight, but calculations for the advance ratio follow that of a conventional puller-type configuration found on propeller-based aircraft.

Figure 15 below shows the normal force coefficient C_{Ty} at different vane deflections across a range of advance ratios from 0 to 0.7 for the propeller-in-wing system. A positive coefficient represents forward vectored thrust created by vane deflection, and a negative coefficient represents the drag of the propeller-vane system embedded in the wing. The bare propeller performance is also plotted for comparison. A consistent difference in thrust vectoring performance with respect to vane angle occurs across advance ratios, with a higher deflection angle generating a more positive C_{Ty} value. At $J < 0.3$, a positively directed normal force may be achieved with at the maximum value of $\delta_v = 30^\circ$. As this is within the bounds of operation for most fixed rotor aircraft, this performance is accepted as realistic

for the anticipated flight conditions for any applied system operation. The results shown in Figure 16 do not account for the drag on the wing airframe which the propeller-vane system needs to overcome to facilitate forward flight. Based on the wind tunnel experimental data collected in UD-LSWT on a sized-down Eppler 479 wing (effective aspect ratio of 4) under similar Reynolds number conditions, the drag coefficient at low angle of attack will be around 0.03. For the wing airframe tested, the magnitude of vectored thrust created by the vanes may thus still overcome the wing-based drag under $J < 0.16$ at a vane tilt angle of 30 degrees.

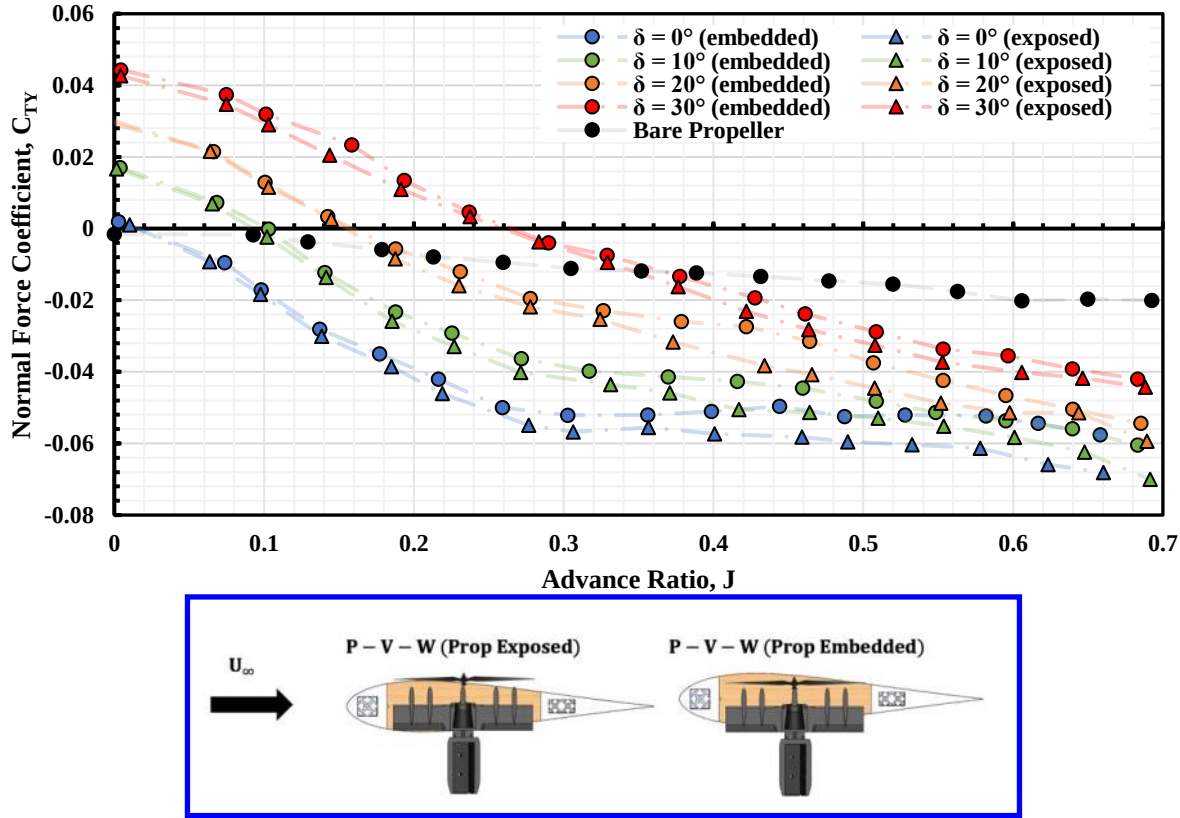


Figure 16. Propeller normal force across changing advance ratio for different vane tilt angle cases.

Drag is an important aspect to consider when determining the efficiency of the vane thrust vectoring system. There are several aspects in the PVW system which contribute to the overall drag coefficient. The vane placement in the wing results in a bleeding through of air from the lower to upper wing surface, subsequently creating drag. The sharp corners at the edge of the wing housing near the vane system lead to flow separation, causing an increase in drag coefficient. The exposed vanes as well as propeller area itself increase the overall drag coefficient of the wing as well. Drag minimization for the PVW system will be the focus of this study as it progresses which will address some of these key issues. However, an unexpected drag component appeared in the PVW system as a result of the propeller spin; significant differences are observed in the drag force between the propeller-on and the propeller-off condition. The direct measurement of drag force ($-F_Y$) of the PVW system at zero vane tilt angle ($\delta_v = 0^\circ$) at different advance ratios is shown in Figure 17. An increment in the drag force corresponds to the increment of advance ratio for all cases, but the drag force experienced by the PVW with propeller-on condition is significantly higher than that of propeller-off, even when comparing the sum of the bare propeller and PWV with propeller-off at $J < 0.52$. As such, the additional drag due to propeller spin is defined as its own distinct phenomena, dubbed propeller-induced drag. Mathematically, the propeller-induced drag coefficient considered here can be described as:

$$C_{Dip} = \frac{D_{Prop\ On} - D_{Prop\ Off}}{\rho n^2 D^4} \quad (6)$$

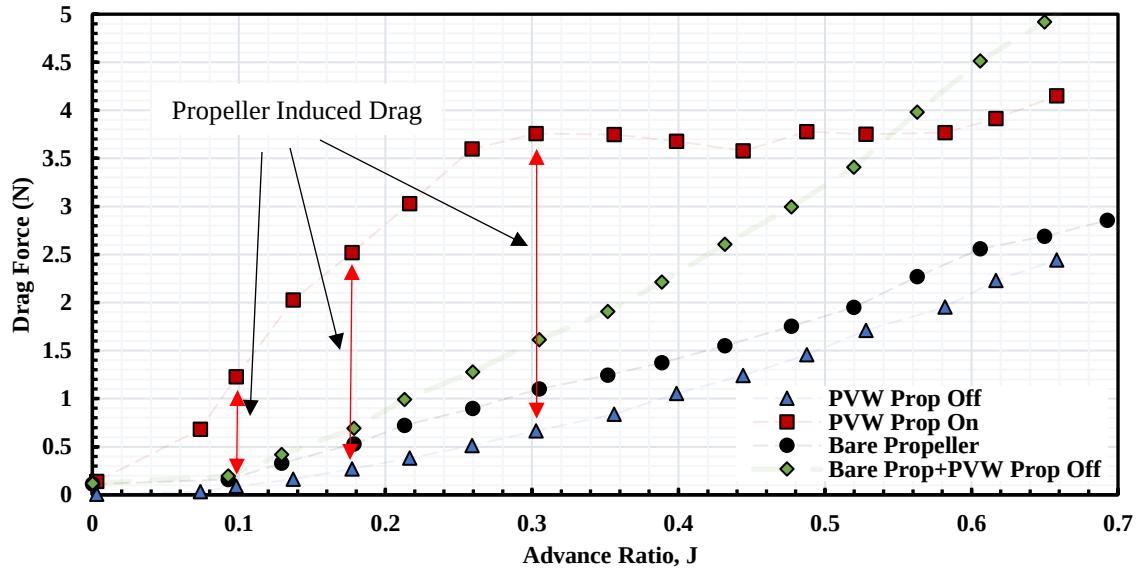


Figure 17. Drag force experienced by different configuration across variable advance ratio conditions, PVW $\delta_v = 0^\circ$ (embedded propeller).

The propeller induced drag coefficient at different vane tilt angles for both PVW configurations, along with the drag expected by the bare propeller operating in the freestream, is highlighted in Figure 18 below. It should be noted once again that the drag coefficient plotted in Figure 18 is purely due to the propeller spin. Negative C_{Dip} in Figure 18 essentially indicates thrust. It can be seen that with increment in advance ratio, the C_{Dip} value for the bare propeller condition increases. Depending on the vane tilt angle, a similar trend is observed in the PVM system for both embedded and exposed propeller; the C_{Dip} increases until a certain advance ratio and plateaus thereafter with increment in advance ratio. Across all the advance ratios, the exposed propeller case showed higher induced drag than the embedded ones, as expected. This difference was magnified at higher advance ratios for all vane tilt angle cases.

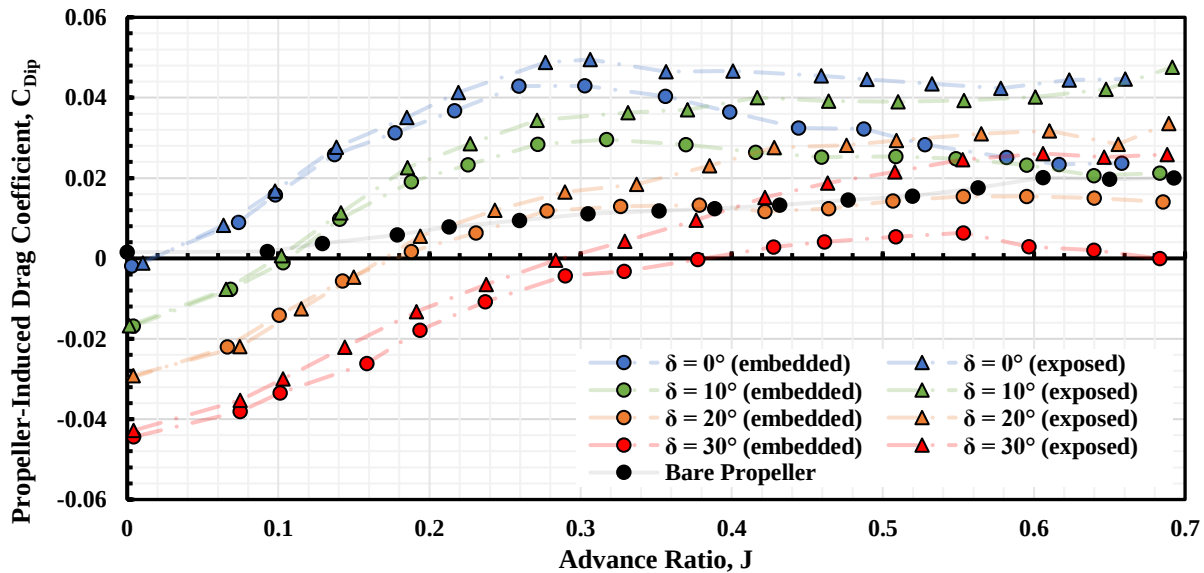


Figure 18. Propeller induced drag across changing advance ratios for different vane tilt angle cases.

Smoke flow visualization was performed on the $\delta_v = 0^\circ$ case at $J = 0.2$ to determine the differences between prop-on and the prop-off case of the PVW system. A smoke wand with a diameter of 1.27 cm was placed upstream of the wing in the tunnel flow. Side views of the PVW system under both conditions are shown in Figure 19. The smoke boundary is traced with the red dashed line in each view. In the propeller-on case, the flow near the lower surface of the wing is sucked into the wing opening from the lower to upper surface and is then pushed downward once again by the propeller. When the propeller is off, the flow visualization shows a different flow behavior. The flow does not cross from the lower surface of the wing to the upper surface. Instead, the flow continues downstream as it normally would across a conventional wing in flight. The deflection and the bending of the freestream in the propeller-on condition is hypothesized to be the additional source of drag experienced by the PVW system. Hence, an improved design of the PVW system in planned subsequent investigations to prevent this unexpected flow bending is essential to provide better forward flight performance for the PVW system.

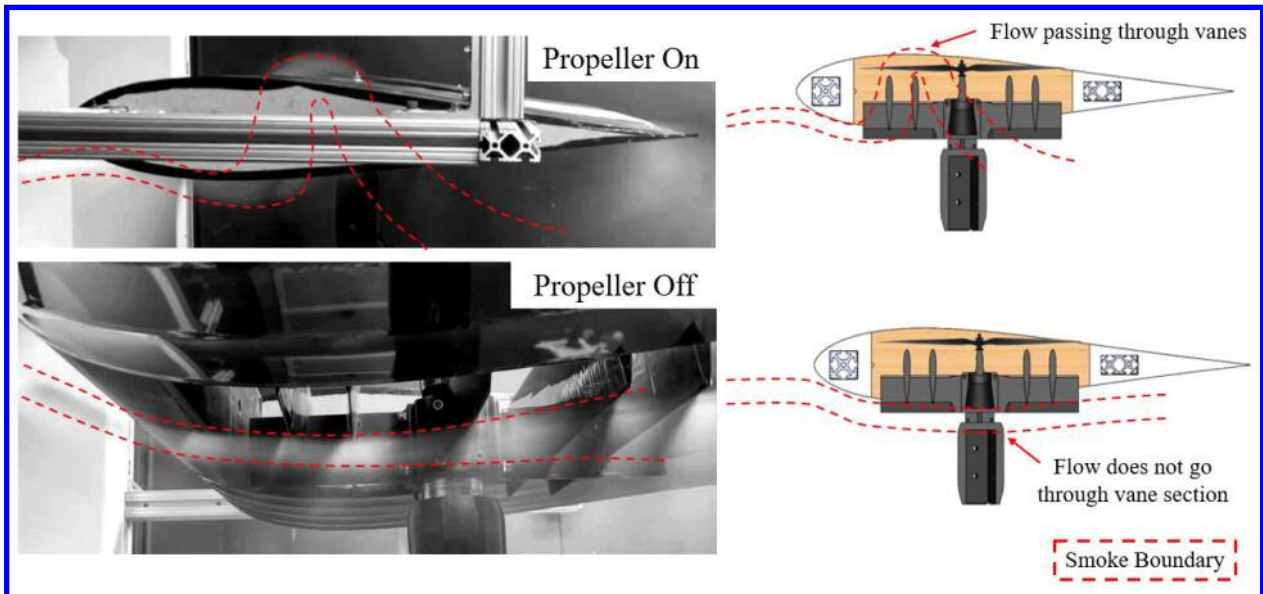


Figure 19. Side and underside views of flow visualization of PVW system (embedded propeller) at $\delta_v = 0^\circ$ and $J = 0.2$.

Behavior of the pitching moment coefficient C_{Qx} for dynamic testing of the propeller-in-wing system is shown below in Figure 20. Negative C_{Qx} indicates a pitch down moment and positive C_{Qx} indicates an upward pitching moment. At $J < 0.15$, stabilizing, negative values of C_{Qx} may be achieved by adjusting the value of δ_v within the experimental bounds of 0 and 30-degree deflection. At $J > 0.15$, however, additional control surfaces or small balancing rotors would be required to provide longitudinal stability. The propeller location appears have a minor effect on C_{Qx} .

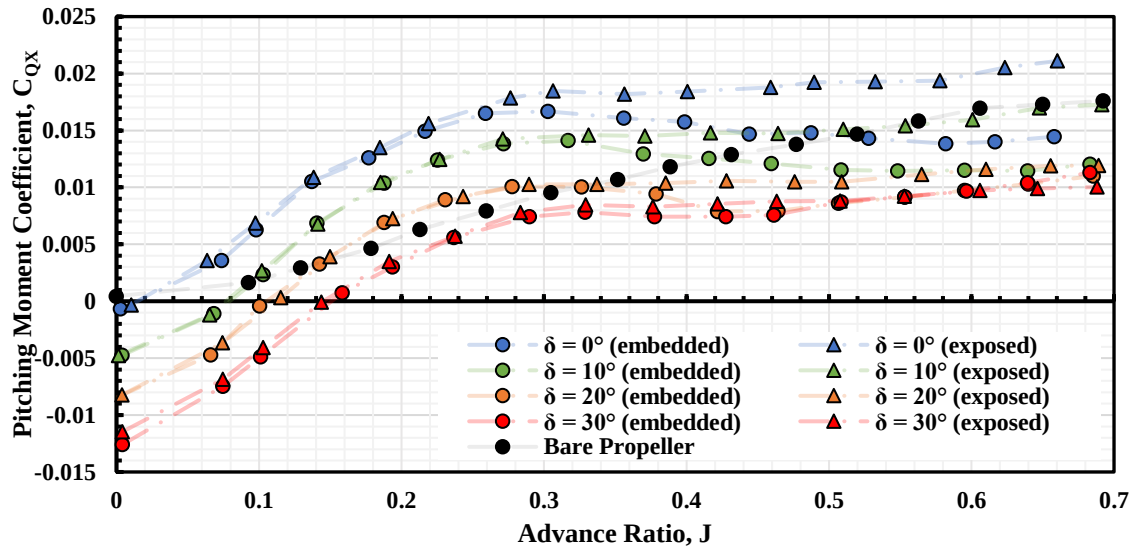


Figure 20. Propeller pitching moment coefficient across advance ratio conditions for different vane tilt angle cases.

Figure 21 below shows the lift-based results of dynamic propeller testing for the integrated wing design. The propeller axial thrust coefficient C_{Tz} increases with changes in advance ratio for both cases. This was also shown in the bare propeller case which further describes the nature of propeller performance when it is configured to a higher angle of attack. At $J < 0.3$, a 4% average difference is observed between the embedded and exposed propeller cases, the exposed cases showing a better performance in C_{Tz} . Recalling thrust vectoring effects, higher vane deflection angle cases exhibit lower thrust coefficient values. Therefore, even though greater thrust is produced by the propeller-in-wing system at higher advance ratios, the ability to vector thrust diminishes significantly due to dominant freestream dynamic pressure compared to that of the propeller wake. As high advance ratio conditions are not realistic for most UAV-type applications, this result does not condemn the system overall. The performance of the $\delta_v = 0^\circ$ and $\delta_v = 10^\circ$ cases shows little disparity in axial thrust despite noticeable changes in normal force.

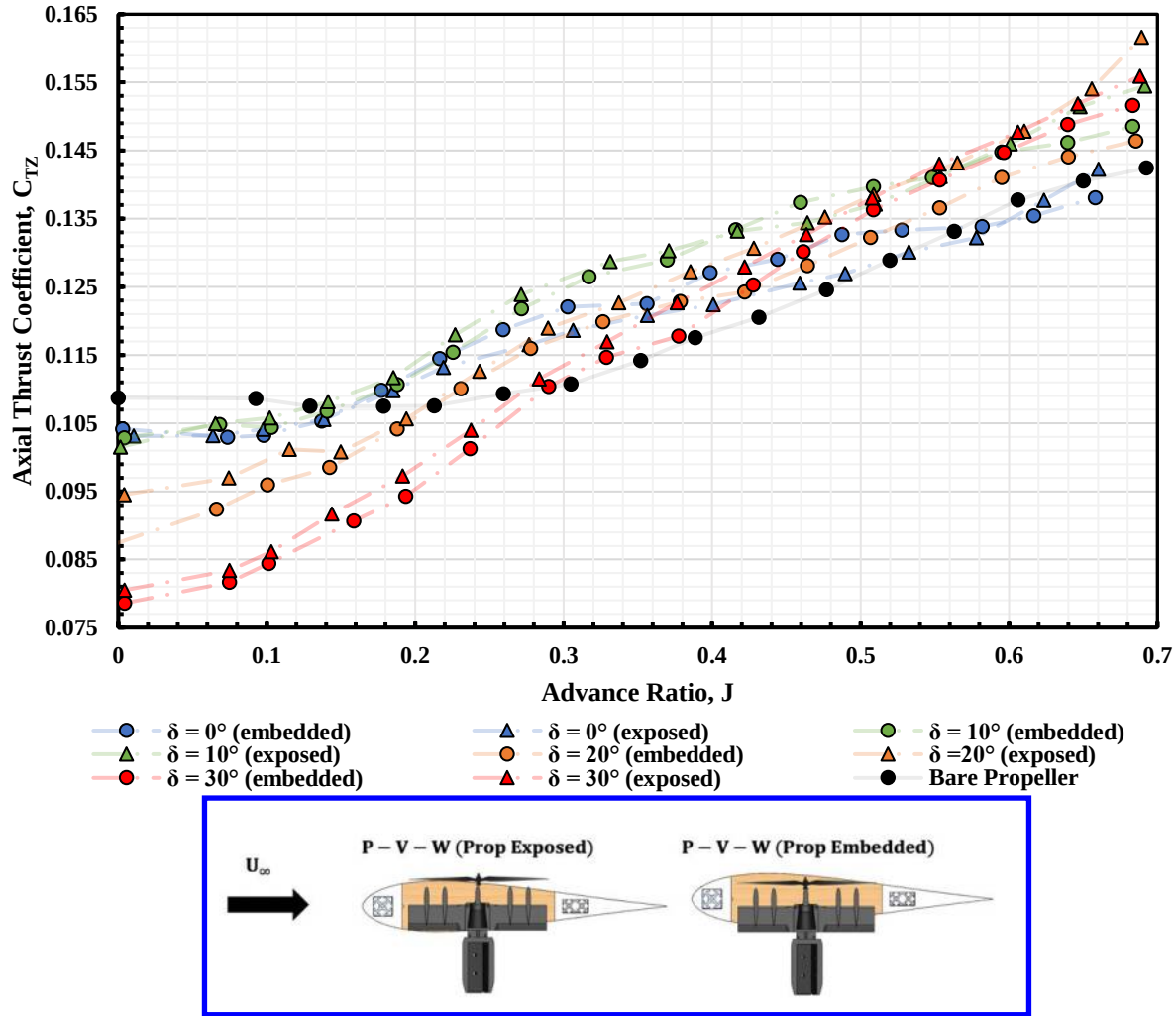


Figure 21. Propeller axial thrust coefficient across advance ratio conditions for different vane tilt angle cases.

The overall efficiency of the propeller vane system in forward flight is quantified by Equation 7. Utilizing the propeller vane system efficacy η_δ (Equation 4), we define the PVW system thrust vectoring efficiency in forward flight as:

$$\eta_{TV} = \eta_\delta * \eta_p \quad (7)$$

where:

$$\eta_p = \frac{C_{P \text{ Bare Prop}}}{C_{P \text{ PVW}}} \quad (8)$$

Figure 22 shows the overall thrust vectoring efficiency for different vane tilt angles. At an advance ratio lower than 0.3, the efficiency eventually drops below 0 even for a 30-degree vane tilt angle. This once again reflects upon the practical advance ratio operating range for the PVW system, which agrees with the results depicted in Figure 16. In general, the overall system thrust vectoring efficiency drops with the increment of advance ratio. Comparing the exposed and embedded propeller cases, the PVW system with propeller embedded has 3% higher overall thrust vectoring efficiency on average than that with the propeller exposed. It is noteworthy, however, that the vertical thrust generated by the exposed propeller is around 2% higher than the embedded propeller (see Figure 21). Thus, the embedded propeller PVW system will have a better performance in the forward flight conditions while the exposed propeller PVW system will have a better performance in the transition flight conditions for the integrated wing.

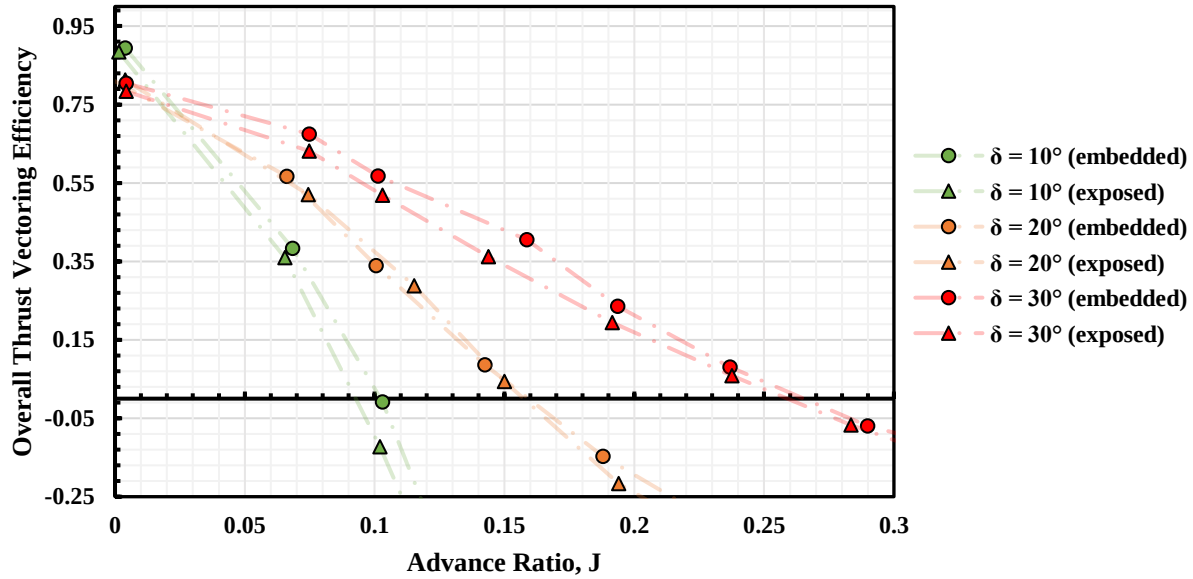


Figure 22. Overall thrust vectoring efficiency across changing advance ratio conditions for different vane tilt angle cases.

V. Conclusions

The efficiency of thrust vectoring capabilities was determined for both a standalone propeller-vane system and a propeller-vane system embedded in a wing (propeller-in-wing system) through both static and wind-on tests. The following are significant takeaways from testing results and analysis:

- Static results indicated that the wing had a positive effect on the propeller-vane system, providing higher thrust vectoring efficiency in hover.
- The propeller-vane system itself was capable of vectoring axial thrust up to 30 degrees while maintaining more than 80% of the overall thrust generated.
- For transition and forward flight, within the practical forward flight speed range where $J < 0.3$, the propeller-in-wing system was able to generate forward thrust; the ability to provide longitudinal stability, however, was limited within $J < 0.15$.
- Propeller location sensitivity analysis revealed that the exposed and embedded propeller test cases for the propeller-in-wing system had similar performance, the exposed featuring a slightly lower thrust vectoring capability but a higher overall efficiency. The exposed PVW system will thus outperform the embedded in hover and transitional flight conditions, but the embedded will perform best in forward flight.

Further study will consider the development of the integrated wing's leading-edge design with a focus on lip geometry to address duct-like performance effects as well as an investigation into the effects of the integrated propeller-vane system at different propeller pitching angles relative to the wing platform.

VI. Acknowledgements

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